E-commerce Sales Analytics:

January 2022

Week 1, Data Analysis project

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Date:

1. Title and Objective

This report analyses e -commerce order data for the first two weeks of January 2022. The objective is to summarize sales performance by product category, region, and payment method, track daily revenue, assess customer satisfaction and delivery performance, and identify recommendations for the business.

2. Dataset Overview

The dataset is

Ecommerce_sales_analytics.5000.csv.Despite its file name, it contains 15 orders placed between 1 and 15 January 2022. Each order has 12 fields: order ID, order date, customer ID, product category, region, quantity, unit price, discount, payment method, delivery days, customer ratings, and revenue.

The orders come from 15 different customers, three product categories (clothing, Beauty, Electronics), four regions (west, south, East, North), and three payment methods (cards, COD, Wallet). The total quantity sold was 78 units.

3. Cleaning summary

In cleaning the dataset, the data contains checks that are required for cleaning purposes mentioned below:

In the above structure, all the numeric fields and order dates were converted from text to date format.

The revenue data contains revenue equal to quantity *unit price*(1 – discount), comprising 15 rows.

The cleaning summary comprises: The unique order ID upon summarising the data set, having a unique value

In the cleaning format, the dataset has no missing values in any column of the dataset

4. KPI summary

The total revenue collected is 51,09,775.80, with a total of 15 orders, an average order value of 2,043,910.296, and a median value of 12,64,074.675. The average discount is 18.8%, the average customer rating is 2.77 out of 5, and the average delivery time is 6.1 days. Orders delivered more than 7 days are 3 out of 15, and orders rated 4 or higher are 3 out of 15.

5. Trend analysis

Clothing is 1531931.72, Beauty is 765860.88, and Electronics is 1829899.22 in revenue. Electronics accounts for the highest 50% of the revenue from the 8 orders, while clothing accounts for 50% of the revenue, and beauty has the highest order value.

Daily revenue changed significantly from January, with 426374.92 on 2 January and the period overall; splitting the period, the second half had 8 orders compared with 7 orders in the first week. The average value trended from 12,36,044 to 20,43,910. This pattern is useful for quarterly data before analysis is done.

6. Customer Insights

Payment method: Card payments generated the maximum revenue from the 7 orders, COD from the 5 orders, and wallet from 3 orders. The average ratings vary sharply by 1.72 for orders with a delivery time of 7.6 days. Card delivery varies from 3.10 days to 4.7 days. Wallet orders vary from 3.73 to 6.7 days. It is the second largest source of revenue.

Product categories: Electronics have the highest product category satisfaction, with 2.27 days, and the slowest delivery is 6.7 days; its average order value is high. The beauty is the highest discount at 26%, which the one of reason its value is high satisfaction is higher 3.08 from the other categories.

Region: The West and South have the highest order count of 12 out of 15 orders. The West has the most orders and a rating of 2.71 and a delivery time of 7 days. The North has the only order in the region for delivery.

Delivery and satisfaction: The delivery time and rating of delivery and rating correlation is 0.14. The explanation for slower delivery is not accounted for by the low-rating sample.

7. Visualization section

The analysis has the following skill visuals, all from the dashboard visualization following are the category are listed:

- **Bar chart (product category by revenue)**. In the product category, Electronics has the highest among the categories and among the category electronics and Beauty have the highest revenue. Beauty leads the average value.
- **Line chart (daily revenue)**. The daily revenue category varies with day-to-day swings, peaking in May among the months.
- **Pie chart. Revenue share by region.** In the revenue category, West has 26%, and North has 24%, for a total of 50%, up from East 24% and South 25% % much smaller.
- **Dashboard mock-up.** Combines the KPI cards with the three cards on a single page.

8. Key findings

From the above findings, we concluded that:

1. In the above category, Electronics has the highest value, and Beauty has the highest average order value.
2. COD has the lowest satisfaction (1.72) and the slowest delivery time (7.6 days); the revenue generated is around 36% of the value in the category.
3. The average order value increased from the second week of the product category amongst the products.
4. Customer satisfaction is low, with an average rating of 2.77and only 20% of the orders rated in the category.
5. In the revenue category, the West and south having the 51% of the revenue in the category in the product category.

9. Recommendation

1. Investigate the COD experience. From the category, review the order handling and communication. In the product category list having the lowest satisfaction is despite being among the strong revenue; some improvements are needed in the offerings section of COD.

2. Protect the clothing in inventory. In the product category required that the keep stocks section be reliable for the category that generates half of the revenue

3. Address home fulfilment. Home has the lowest ratings in the category; further investigation is required for its shipping process before spending in the marketing category, realising the shipping procedure before doing analysis

4. Test discount in the Beauty category: Beauty has the highest order value compared to any other category in the category section; there required to encourage the discount value in the category section.

5. Engaging more data for analysis. In the category, there are fifteen data points recorded in the category, which cannot support the solution; the required dataset is required for the analysis on three months of data or the full dataset

10. Limitation or assumptions

- Sample size: The dataset has 15 orders, which is not 5000 implied by the file name. Percentages and averages that cannot change substantially with a few more orders, mostly in the regions and categories with the three orders.
- Customer ratings: Based on the analysis, the rating is only done for customer purchases on time; there is no time scale for repeating mention are not presented in the category
- Time periods: are not consent the time periods required on the analysis are only 15 days with is not consent on the current discoveries.
- Assumed revenue definition required: Revenue calculated on the quantity and (1- discount) of the purchase, which is not included in the category cost margins, and returns are not included.
- Correlation: The correlation reported on the 15 products in the categories in the treated sample size only.